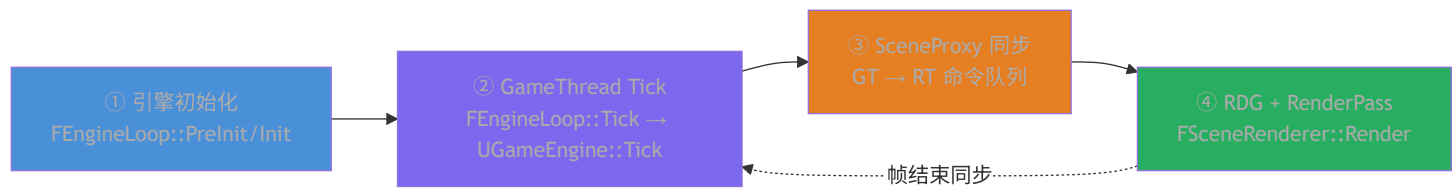


UE5 从引擎初始化到最终渲染的完整管线

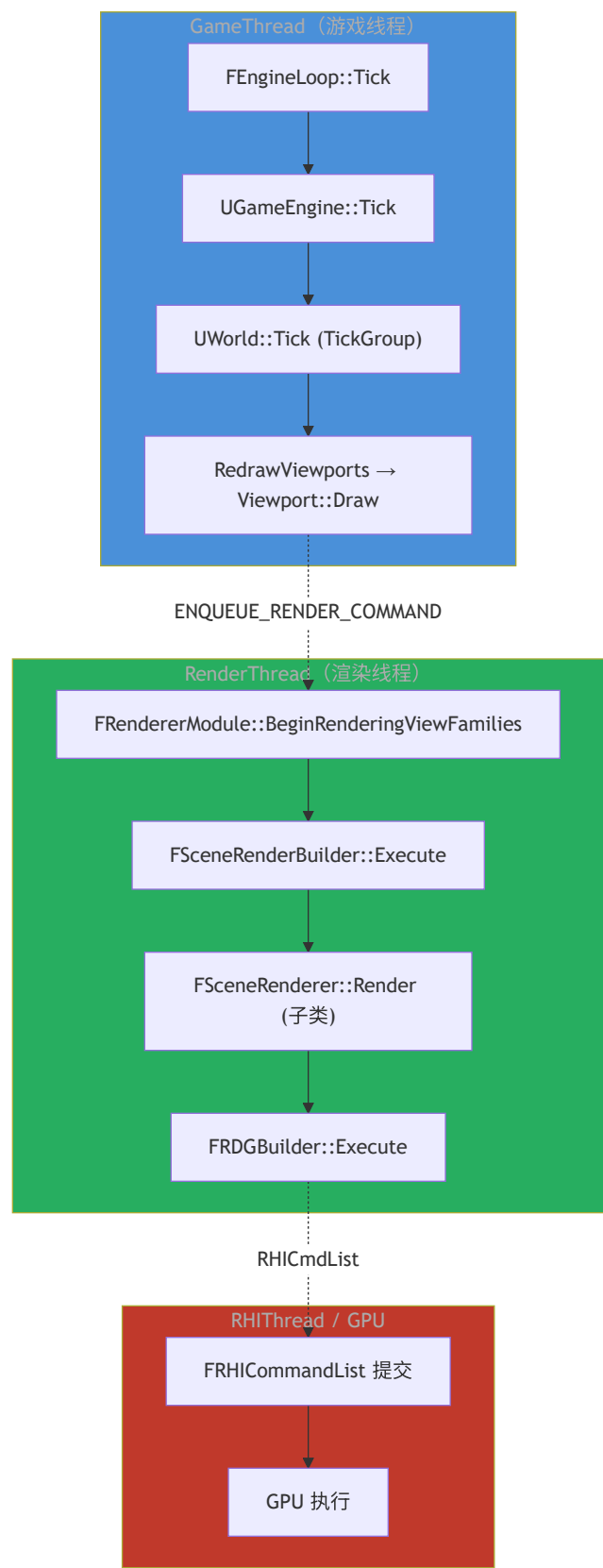
本文档梳理虚幻引擎（UE5）从进程启动 → 引擎初始化 → GameThread 每帧 Tick → SceneProxy 构建/同步 → RDG 与 RenderPass 执行的完整数据流与关键类/函数，所有源码位置基于 `j:\UnrealSource\UnrealEngine`。文中所有文件引用均可点击跳转。

0. 全局架构总览

0.1 四大阶段一览



0.2 多线程协作模型

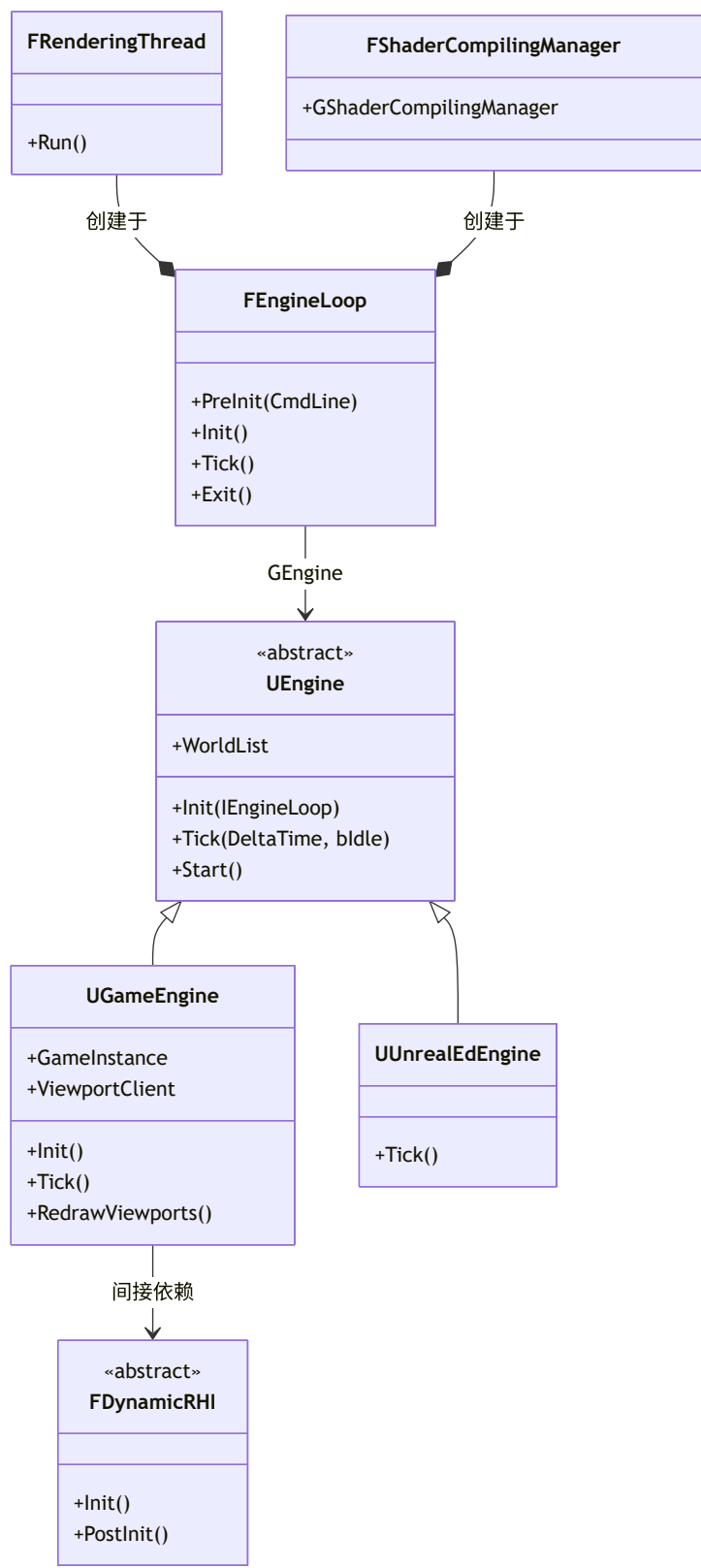


0.3 阶段职责速查

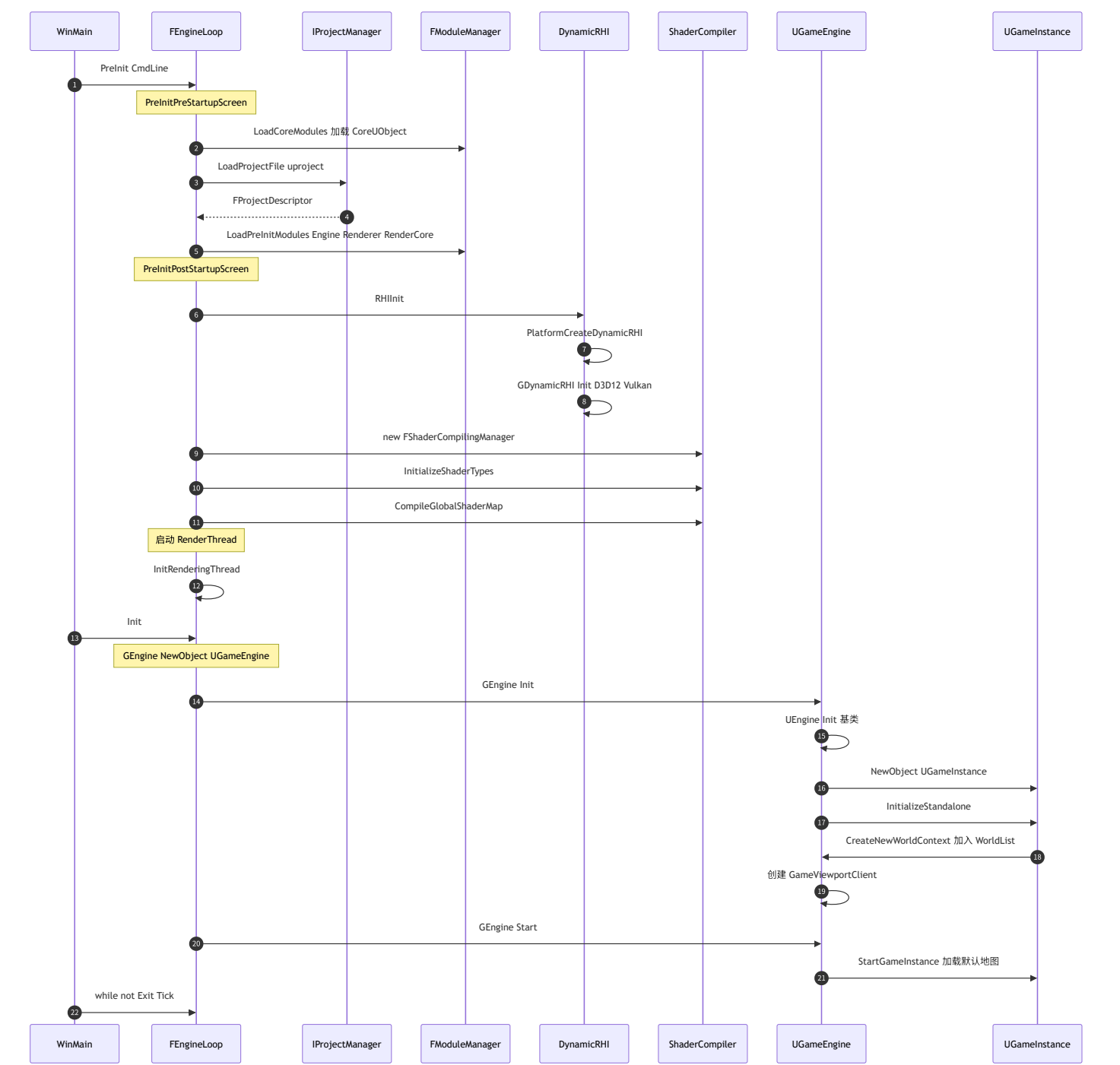
阶段	线程	关键入口
① 初始化	MainThread	[FEngineLoop::Init] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4682)
② Tick	GameThread	[FEngineLoop::Tick] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5536)
③ SceneProxy 同步	GT→RT	[UWorld::SendAllEndOfFrameUpdates] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/LevelTick.cpp#L1346)
④ 渲染	RenderThread	[FDeferredShadingSceneRenderer::Render] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/DeferredShadingRenderer.cpp#L1736)

1. 引擎初始化阶段

1.1 关键类关系



1.2 初始化时序图



1.3 入口与启动流程

函数/对象	文件	行号	
WinMain	[LaunchWindows.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/Windows/LaunchWindows.cpp#L315)	315-337	WinMain 入口
GuardedMain	[Launch.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/Launch.cpp#L87)	87-205	GuardedMain 启动流程
GEngineLoop (全局)	[Launch.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/Launch.cpp#L28)	28	GEngineLoop 全局函数

函数/对象	文件	行号	
class FEngineLoop	[LaunchEngineLoop.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Public/LaunchEngineLoop.h#L48)	48- 200	弓

1.4 FEngineLoop 关键函数

函数	文件	行号	
PreInit(const TCHAR*)	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4332)	4332- 4359	尸
PreInitPreStartupScreen	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L1699)	1699- 3385	角
PreInitPostStartupScreen	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3387)	3387- 4331	尸
LoadCoreModules	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4361)	4361- 4371	仇
LoadPreInitModules	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4379)	4379- 4432	E 冬
LoadStartupCoreModules	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4438)	4438- 4601	~
LoadStartupModules	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4604)	4604- 4630	拊
Init	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4682)	4682- 4908	仓
Tick	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5536)	5536- 6168	套

1.5 UEngine / UGameEngine 关键函数

函数	文件	行号
GEngine 赋值点	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4714)	4714
GEngine->Init 调用点	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L4763)	4763
UEngine::Init	[UnrealEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/UnrealEngine.cpp#L2052)	2052
UGameEngine::Init	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L1205)	1205 1315
UGameEngine::Start	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L1317)	1317
GameInstance 创建	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L1249)	1249
CreateNewWorldContext (声明)	[Engine.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/Engine.h#L3558)	3558

1.6 RHI 初始化

函数/对象	文件
RHIInit 调用点	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3111)
RHIInit 实现	[DynamicRHI.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RHI/Private/DynamicRHI.cpp#L278)
PlatformCreateDynamicRHI (Win)	[WindowsDynamicRHI.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RHI/Private/Windows/WindowsDynamicRHI.cpp#L1243)
GDynamicRHI (定义)	[DynamicRHI.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RHI/Private/DynamicRHI.cpp#L35)

注意：RHI 初始化位于 PreInit 阶段（不是 Init 阶段），在 Shader 系统初始化之前完成。

1.7 渲染线程启动

函数	文件	行号	
InitRenderingThread	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L795)	795-807	U
StartRenderingThread	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L562)	562-654	真
class FRenderingThread	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L324)	324	R F
ShutdownRenderingThread	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L809)	809	停
GUseThreadedRendering	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L49)	49	渲

1.8 Shader 系统

函数/对象	文件
GShaderCompilingManager = new	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3191)
InitializeShaderTypes	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3235)
CompileGlobalShaderMap	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3247)
FShaderCompilingManager	[ShaderCompiler.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/ShaderCompiler/ShaderCompiler.cpp#L74)
FShaderType	[Shader.h](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/Shader.h#L1237)

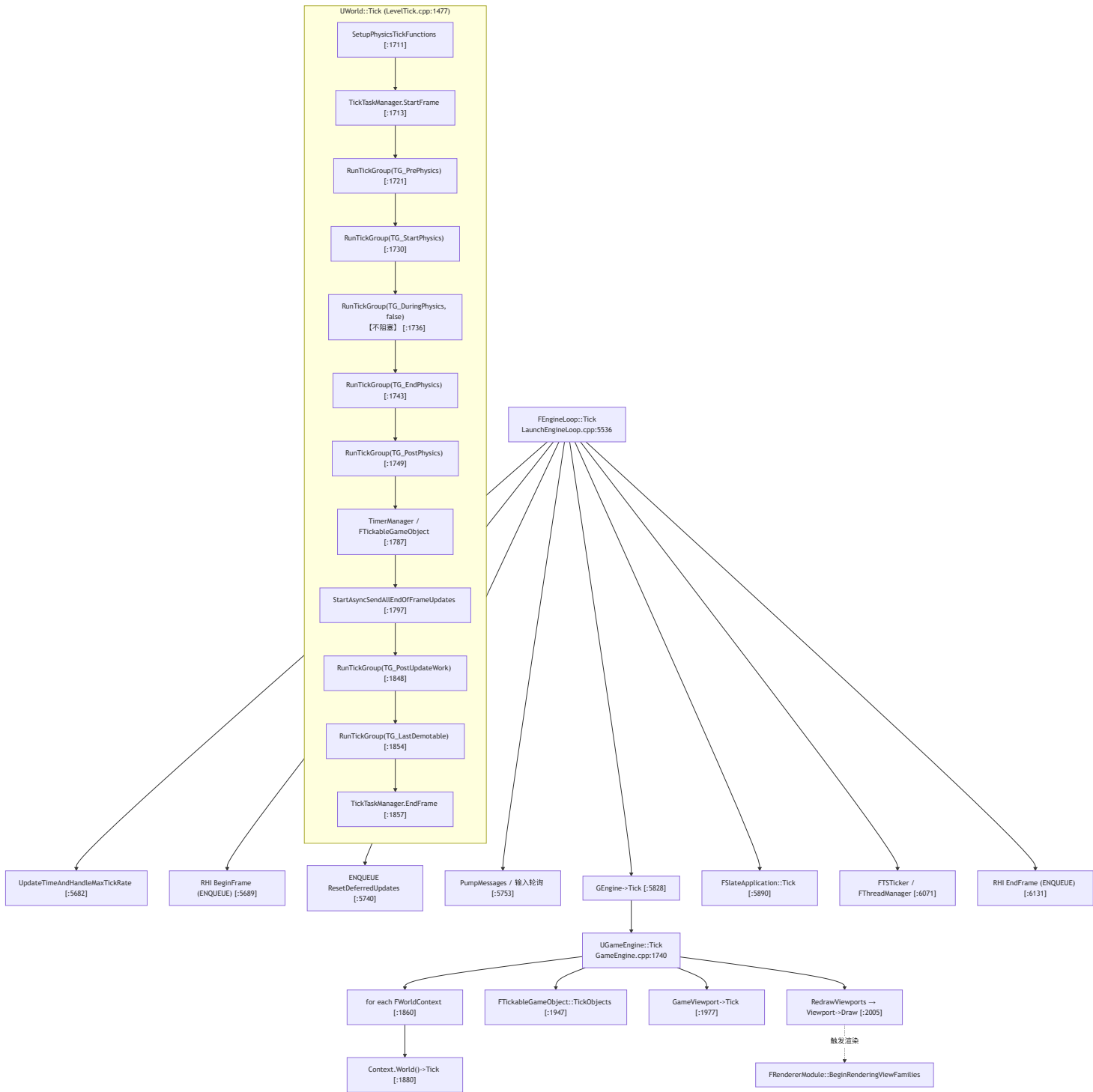
1.9 关键全局对象

全局对象	类型	定义位置
GEngine	UEngine*	[UnrealEngine.cpp:427] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/UnrealEngine.cpp#L427)
GWorld	UWorldProxy	[World.cpp:748] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/World.cpp#L748)

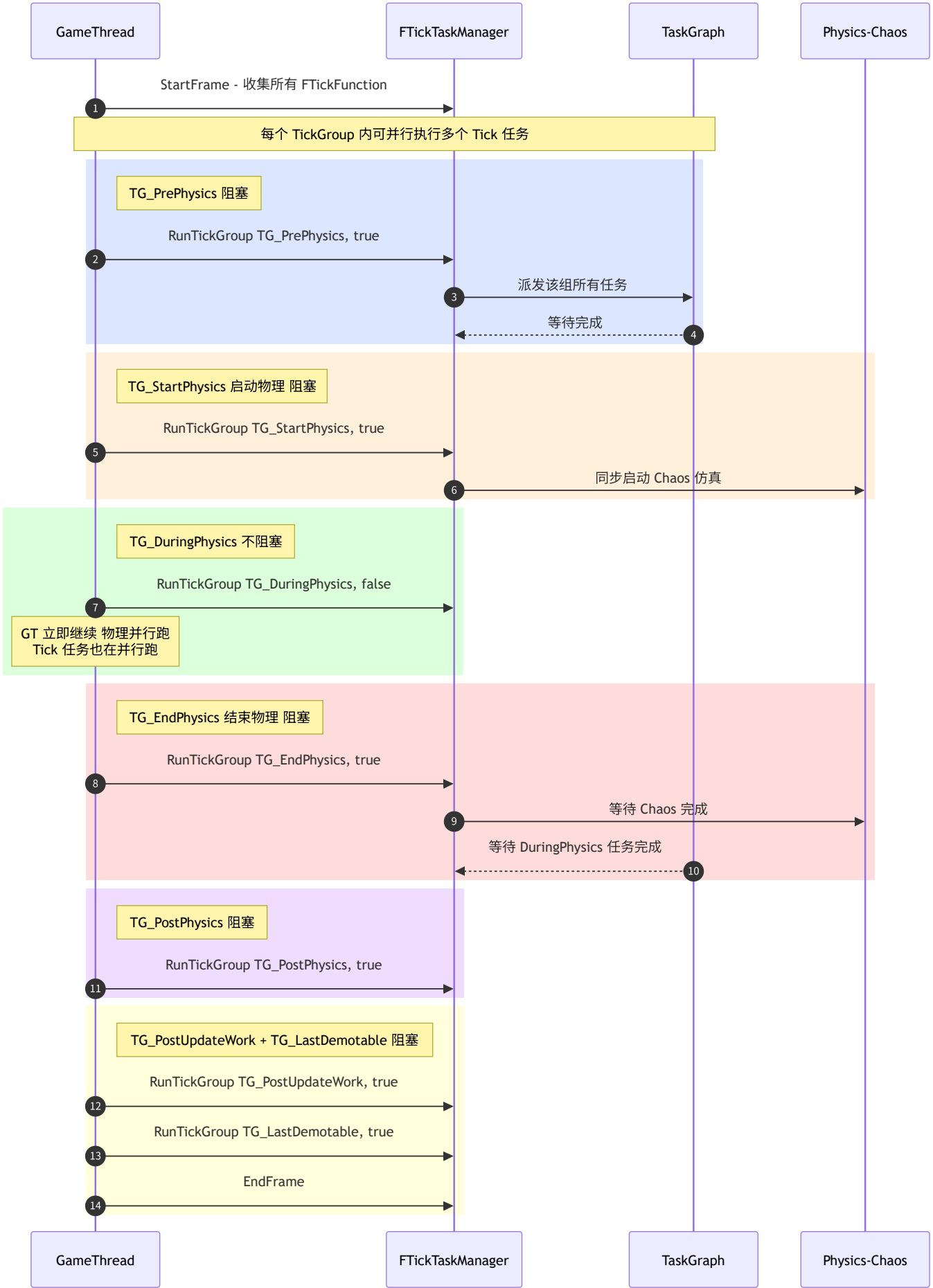
全局对象	类型	定义位置
GDynamicRHI	FDynamicRHI*	[DynamicRHI.cpp:35] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RHI/Private/DynamicRHI.cpp#L35)
GUseThreadedRendering	bool	[RenderingThread.cpp:49] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cp
GThreadPool	FQueuedThreadPool*	[ThreadingBase.cpp:48] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Core/Private/HAL/ThreadingBase.cpp#L4
GEngineLoop	FEngineLoop	[Launch.cpp:28] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/Launch.cpp#L28)

2. GameThread Tick 与 WorldTick

2.1 Tick 函数调用链



2.2 TickGroup 顺序与因果性（重点）



2.3 ETickingGroup 枚举

枚举值	行号	语义
TG_PrePhysics	[EngineBaseTypes.h:86] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L86)	物理仿真开始前
TG_StartPhysics	[EngineBaseTypes.h:89] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L89)	启动物理（Hidden）
TG_DuringPhysics	[EngineBaseTypes.h:92] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L92)	与物理并行
TG_EndPhysics	[EngineBaseTypes.h:95] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L95)	结束物理（Hidden）
TG_PostPhysics	[EngineBaseTypes.h:98] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L98)	物理完成后
TG_PostUpdateWork	[EngineBaseTypes.h:101] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L101)	更新工作完成后
TG_LastDemotable	[EngineBaseTypes.h:104] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L104)	降级 Tick 兜底（Hidden）
TG_NewlySpawned	[EngineBaseTypes.h:107] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L107)	新 spawned（每帧末反复重绘）

2.4 Tick 并行化核心类

注意：UE5 中**不存在** FTickFunctionManager / ULevel::Tick，实际承担职责的是 FTickTaskManager + FTickTaskLevel，FDeferredUpdateInterface 实为 FDeferredUpdateResource。

函数/类	文件
ETickingGroup 枚举	[EngineBaseTypes.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L83)
FTickFunction	[EngineBaseTypes.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Engine/EngineBaseTypes.h#L187)
FTickTaskManagerInterface	[TickTaskManagerInterface.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/TickTaskManagerInterface.h#L18)
FTickTaskManager	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L1965)
FTickTaskManager::StartFrame	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L2000)
FTickTaskManager::RunTickGroup	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L2115)
FTickTaskSequencer	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L456)
FTickTaskSequencer::ReleaseTickGroup	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L933)
FTickFunction::QueueTickFunction	[TickTaskManager.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L2617)

2.5 关键 Tick 函数

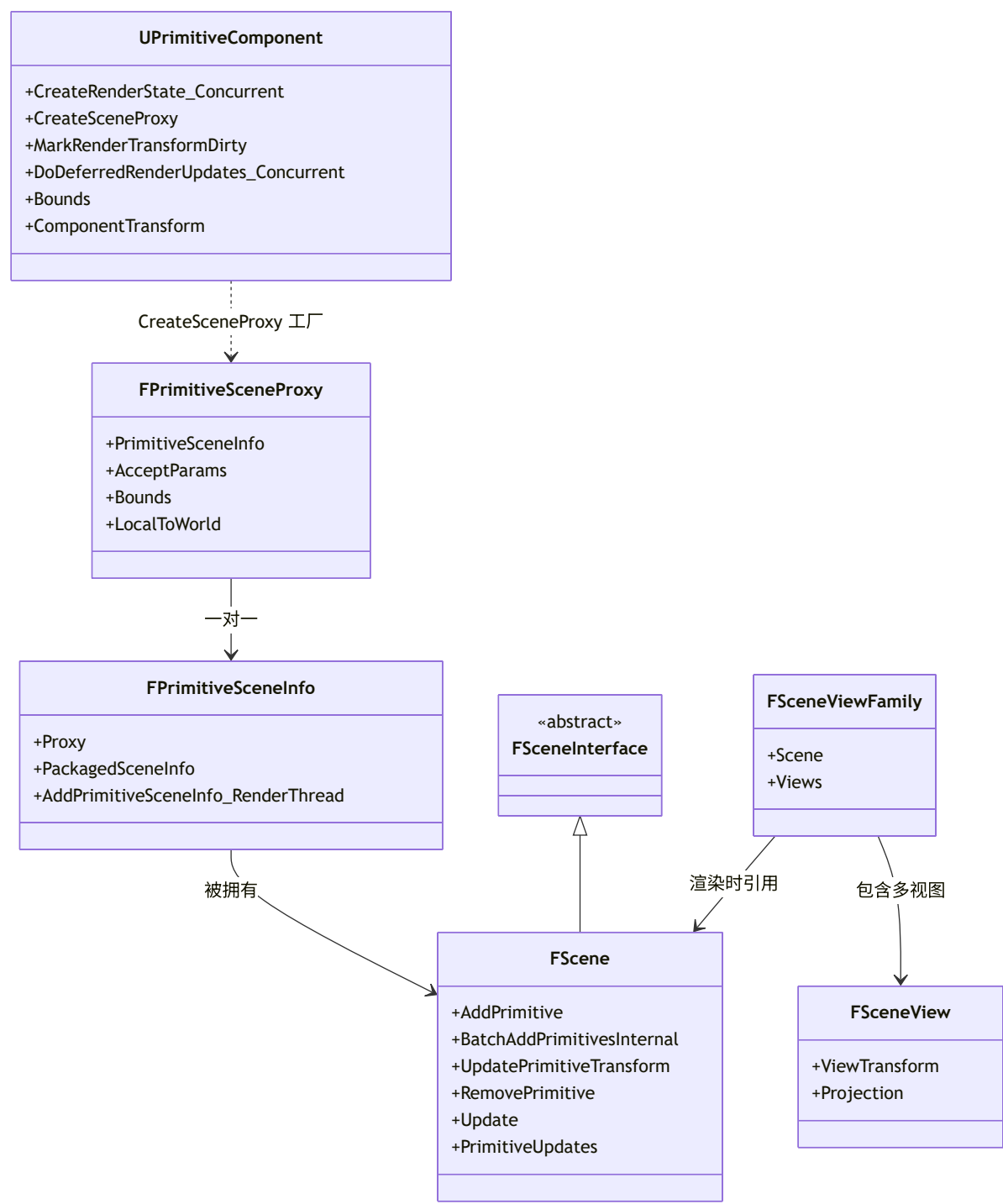
函数	文件	行号	
FEngineLoop::Tick	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5536)	5536-6168	引擎每帧
GEngine->Tick 调用	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5828)	5828	派发到 UGameEngine
UGameEngine::Tick	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L1740)	1740	遍历 World 每个 World
UWorld::Tick	[LevelTick.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/LevelTick.cpp#L1477)	1477-1960	单 World
AActor::Tick	[Actor.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Actor.cpp#L1971)	1971	仅蓝图 Actor + Latent

2.6 GT → RT 连接点

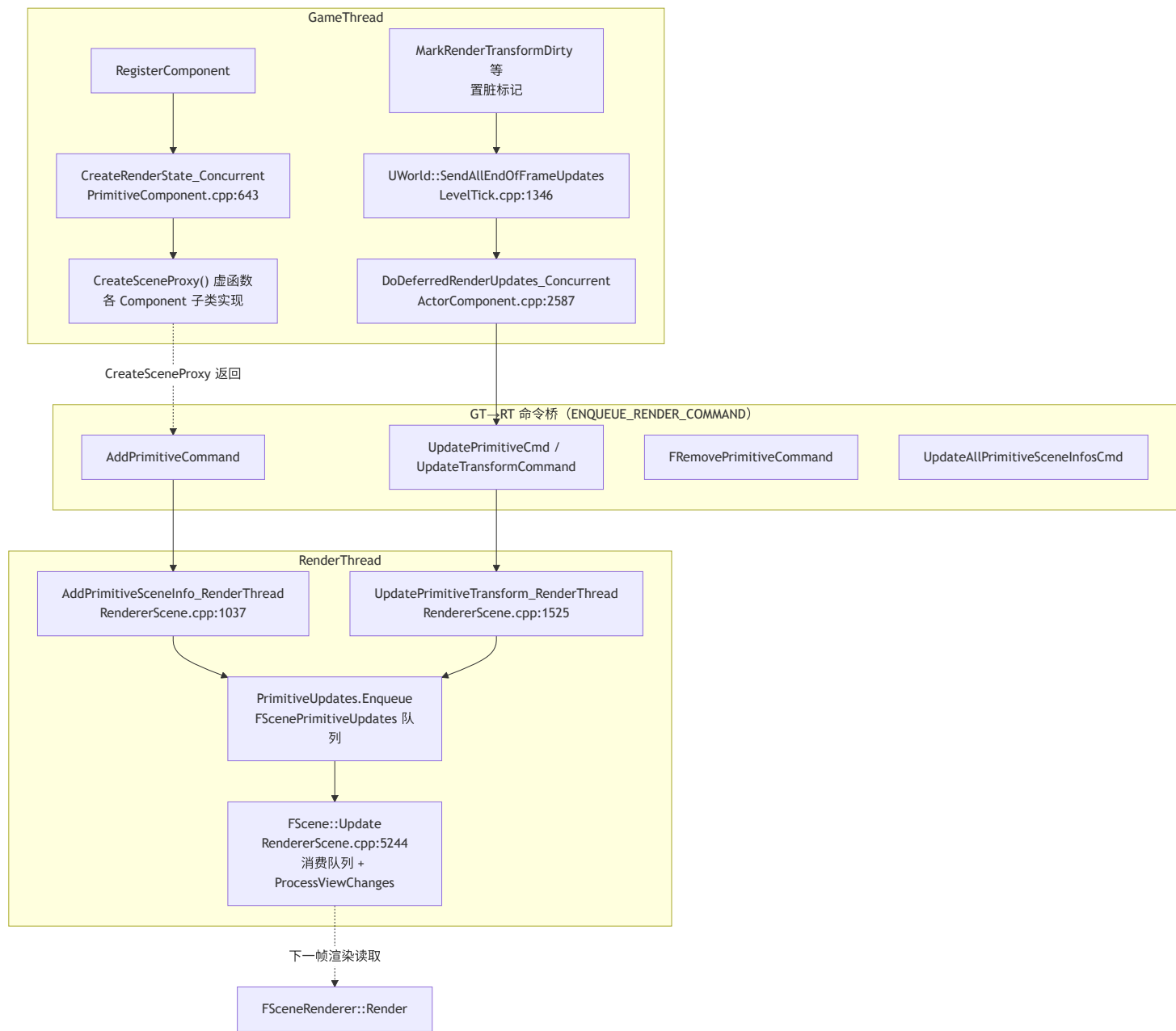
连接点	文件	行号	
RHI BeginFrame	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5689)	5689-5692	引擎每帧
FScene::StartFrame	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5694)	5694-5700	每帧
ResetDeferredUpdates	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L5740)	5740-5744	帧开始
RHI EndFrame	[LaunchEngineLoop.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L6131)	6131-6135	引擎每帧
RedrawViewports	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L2005)	2005	帧结束
RedrawViewports 实现	[GameEngine.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/GameEngine.cpp#L775)	775	调用
FDeferredUpdateResource	[TextureResource.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/TextureResource.h#L285)	285	延迟帧

3. SceneProxy 构建与同步

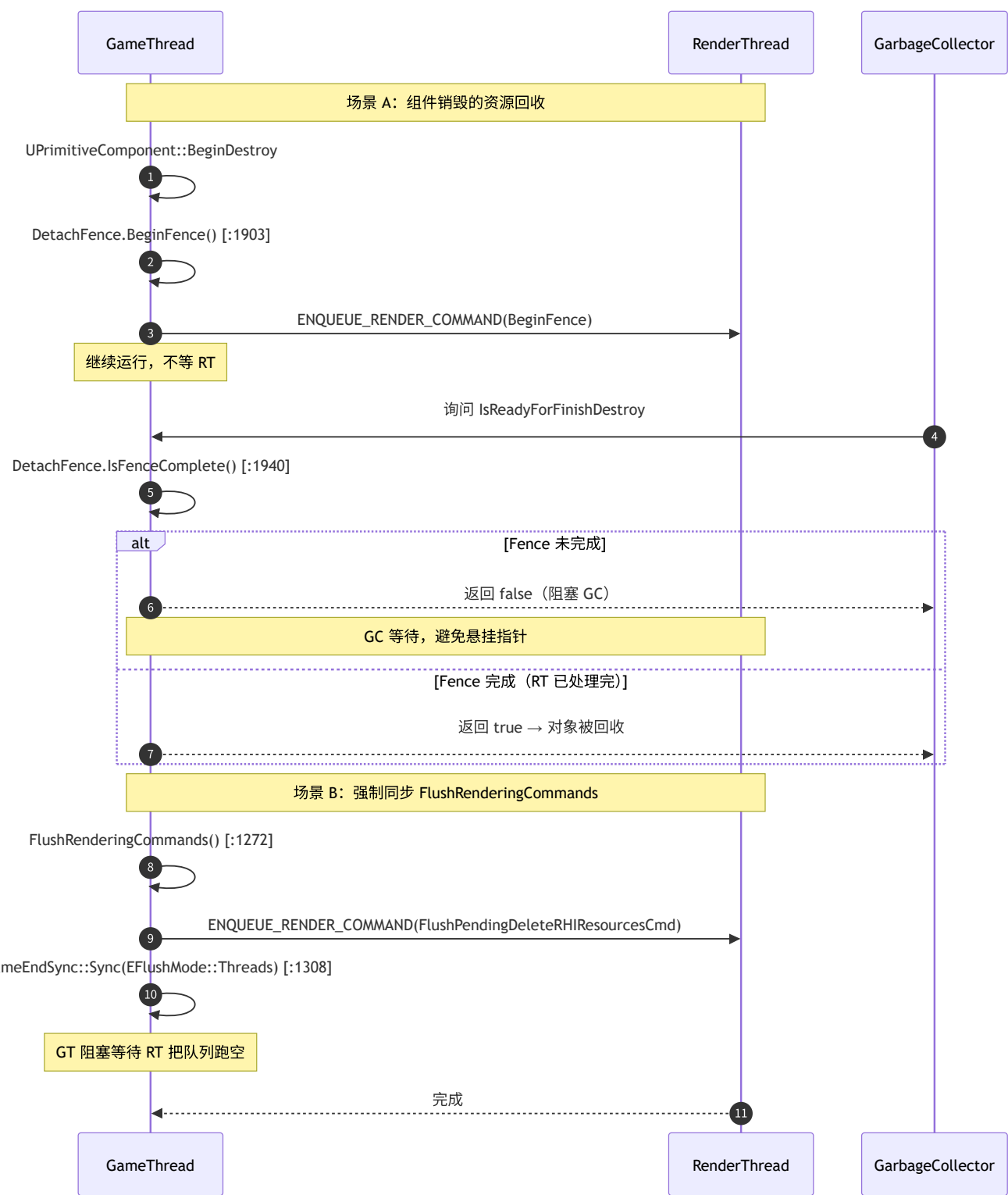
3.1 类关系图（GT 与 RT 的镜像）



3.2 SceneProxy 生命周期与同步机制



3.3 RenderCommandFence 同步机制



3.4 SceneProxy 核心类

类	文件	行号	
FPrimitiveSceneProxy	[PrimitiveSceneProxy.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/PrimitiveSceneProxy.h#L295)	295	UPrimitive的RT镜
FPrimitiveSceneInfo	[PrimitiveSceneInfo.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Public/PrimitiveSceneInfo.h#L265)	265	SceneProc的渲染器

类	文件	行号	
FScene	[ScenePrivate.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/ScenePrivate.h#L2874)	2874	渲染器私有 (UWorld)
FSceneView	[SceneView.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/SceneView.h#L1423)	1423	单个视图 屏幕投影
FSceneViewFamily	[SceneView.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/SceneView.h#L2210)	2210	共享 Scene 的视图集
FSceneInterface	[SceneInterface.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/SceneInterface.h)	—	FScene 的

3.5 SceneProxy 创建链路

函数	文件
UPrimitiveComponent::CreateRenderState_Concurrent	[PrimitiveComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/PrimitiveComponent.cpp)
UPrimitiveComponent::CreateSceneProxy	[PrimitiveComponent.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Classes/Components/PrimitiveComponent.h)
UStaticMeshComponent::CreateSceneProxy	[StaticMeshSceneProxy.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/StaticMeshSceneProxy.cpp)
USkinnedMeshComponent::CreateSceneProxy	[SkinnedMeshComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/SkinnedMeshComponent.cpp)
UInstancedStaticMeshComponent::CreateSceneProxy	[InstancedStaticMesh.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/InstancedStaticMeshComponent.cpp)
ULandscapeComponent::CreateSceneProxy	[Landscape.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Landscape/Private/LandscapeComponent.cpp)
FScene::AddPrimitive	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp)
FScene::BatchAddPrimitivesInternal	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp)
FScene::AddPrimitiveSceneInfo_RenderThread	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp)

3.6 每帧 SceneProxy 更新

函数	文件
UWorld::SendAllEndOfFrameUpdates	[LevelTick.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/LevelTick.cpp)
UWorld::SendAllEndOfFrameUpdatesInternal	[LevelTick.cpp](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/LevelTick.cpp)
UActorComponent::DoDeferredRenderUpdates_Concurrent	[ActorComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/ActorComponent.cpp)
UActorComponent::RecreateRenderState_Concurrent	[ActorComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/ActorComponent.cpp)
UActorComponent::MarkRenderStateDirty	[ActorComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/ActorComponent.cpp)

函数	文件
UPrimitiveComponent::SendRenderTransform_Concurrent	[PrimitiveComponent.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/Components/Primi
FScene::UpdatePrimitiveTransformInternal	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.
FScene::UpdatePrimitiveTransforms	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.
FScene::UpdatePrimitiveTransform_RenderThread	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.

3.7 Scene 更新命令队列（数据结构层）

文件 [ScenePrimitiveUpdates.h](file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/ScenePrimitiveUpdates.h) 定义了：

- EPrimitiveUpdateDirtyFlags （行 14-37）： Transform/InstanceData/CullingBounds 等脏标记
- EPrimitiveUpdateId （行 39-52）： UpdateTransform / UpdateInstance / CustomPrimitiveData / DrawDistance 等
- FUpdateTransformCommand （行 61-68）、FUpdateInstanceCommand （行 70-76）等载荷
- FScenePrimitiveUpdates （行 54）： TSceneUpdateCommandQueue<FPrimitiveSceneInfo, ...> ，所有增删改统一队列

3.8 RenderCommandFence 同步

项	文件	
class FRenderCommandFence	[RenderCommandFence.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderCommandFence.h#L14)	
enum ESyncDepth	[RenderCommandFence.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderCommandFence.h#L17)	
FRenderCommandFence::BeginFence	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L978)	
FRenderCommandFence::Wait	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L1258)	
FlushRenderingCommands	[RenderingThread.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L1272)	

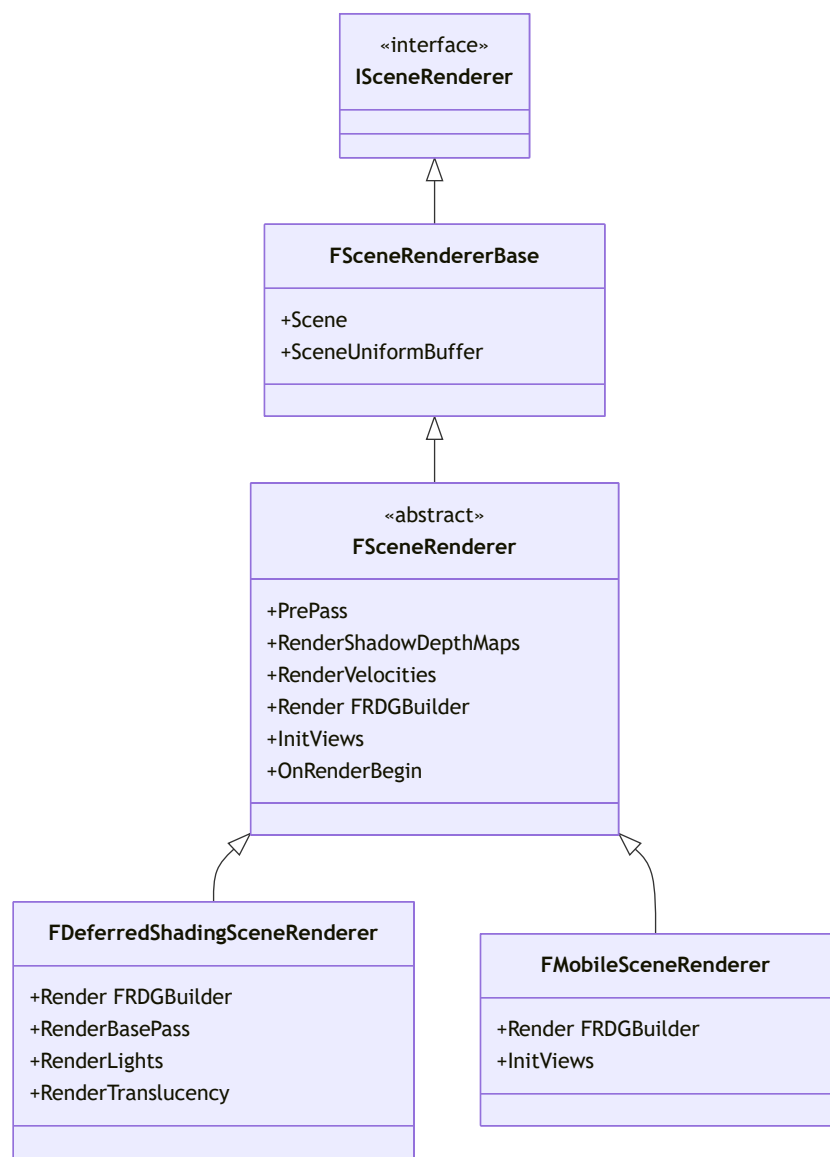
本代码库中**不存在** FlushRendering （无后缀）和 StartRenderCommandFence ，最接近的是 FRenderCommandFence::BeginFence 与 StartRenderCommandFenceBundler （[LaunchEngineLoop.cpp:3832] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Launch/Private/LaunchEngineLoop.cpp#L3832)， GC 期间优化用）。

3.9 整场景重建（FScene::Update）

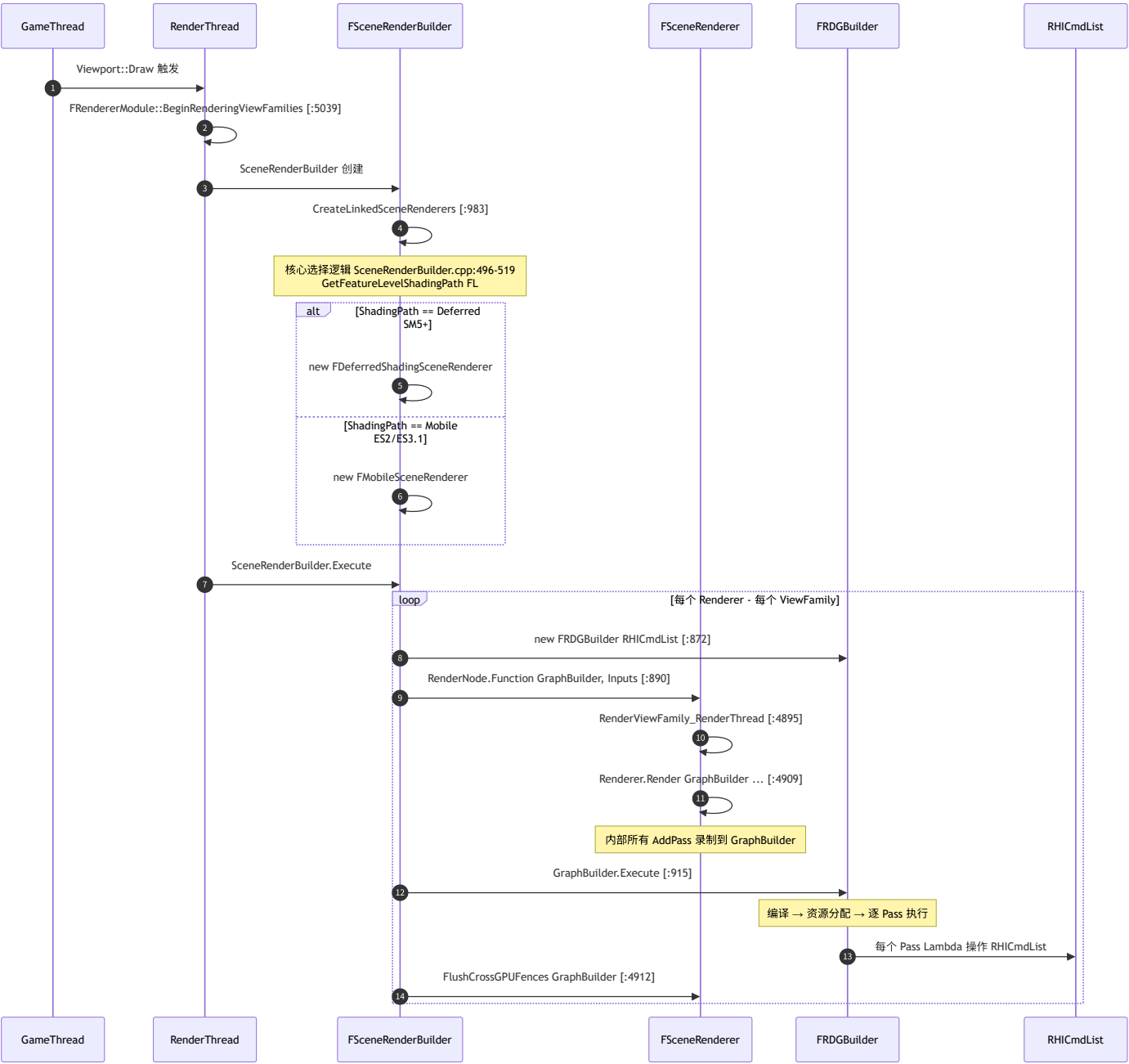
函数	文件
FScene::UpdateAllPrimitiveSceneInfos	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp#L515)
FScene::Update	[RendererScene.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp#L524)
FGlobalComponentRecreateRenderStateContext	[ComponentRecreateRenderStateContext.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/ComponentRecreateRenderStateContext.h#L10)

4. RDG 与 RenderPass 执行

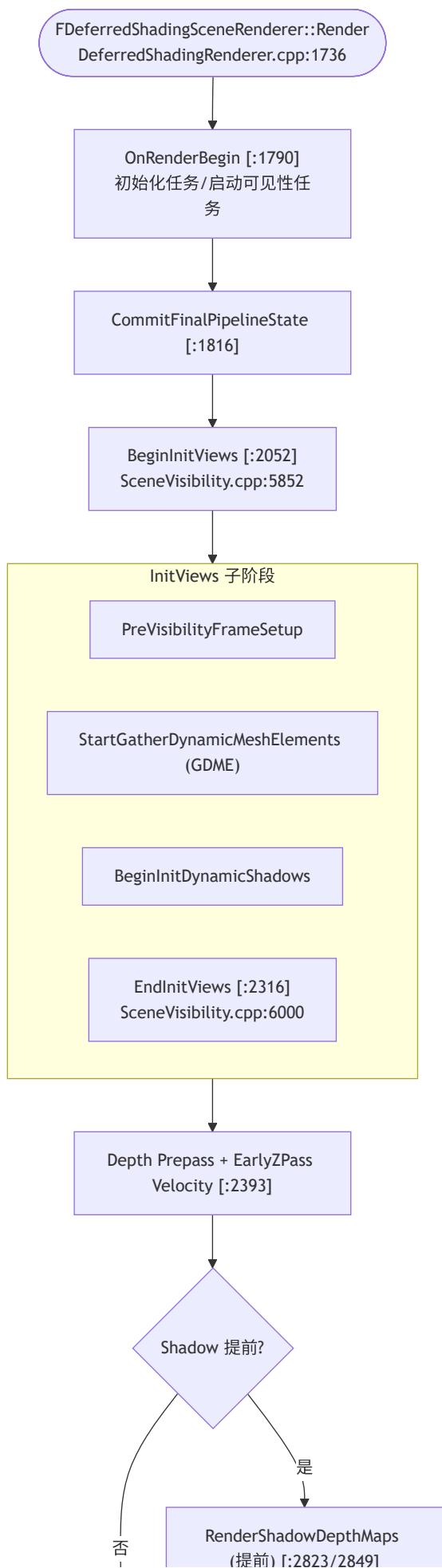
4.1 渲染器继承体系

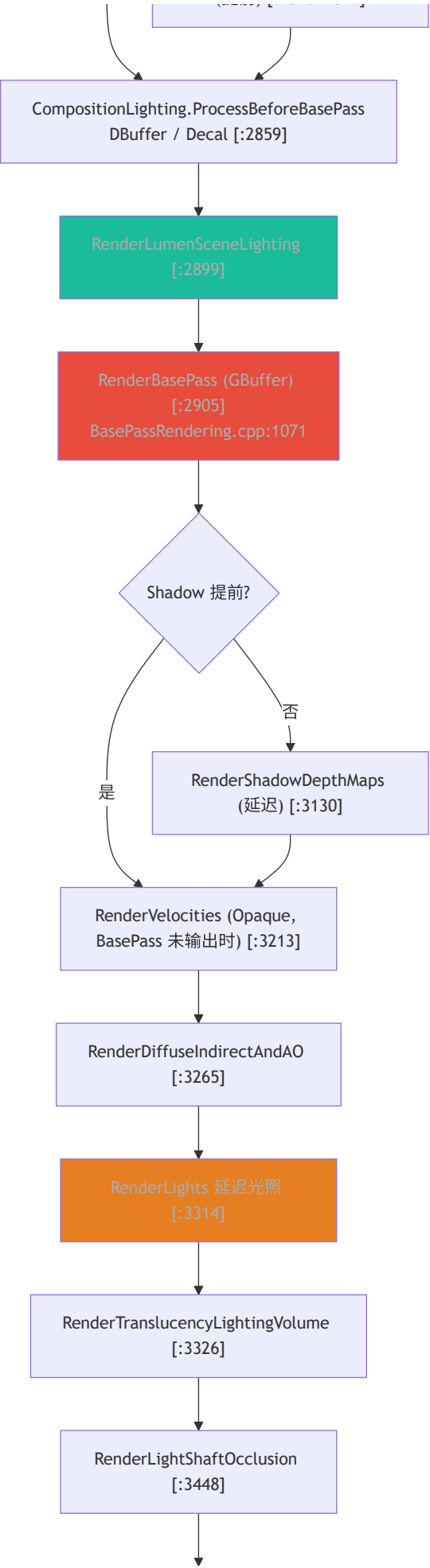


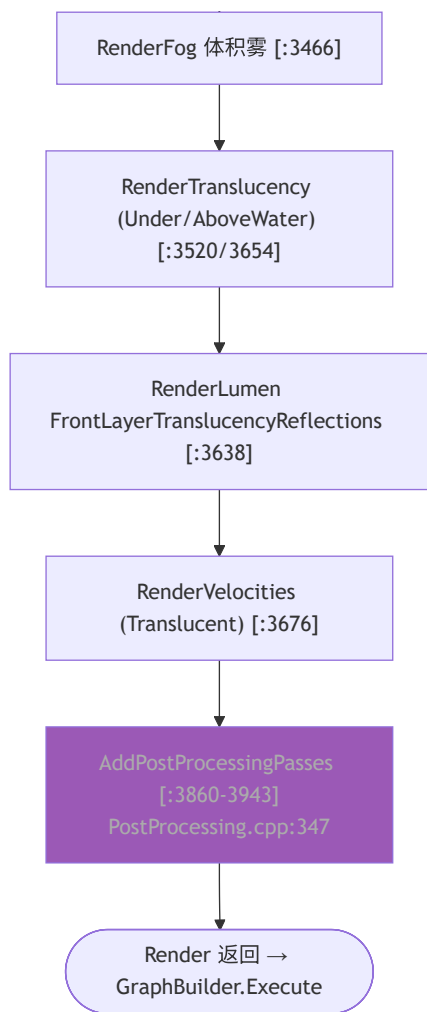
4.2 渲染入口 → Renderer 选择 → Render



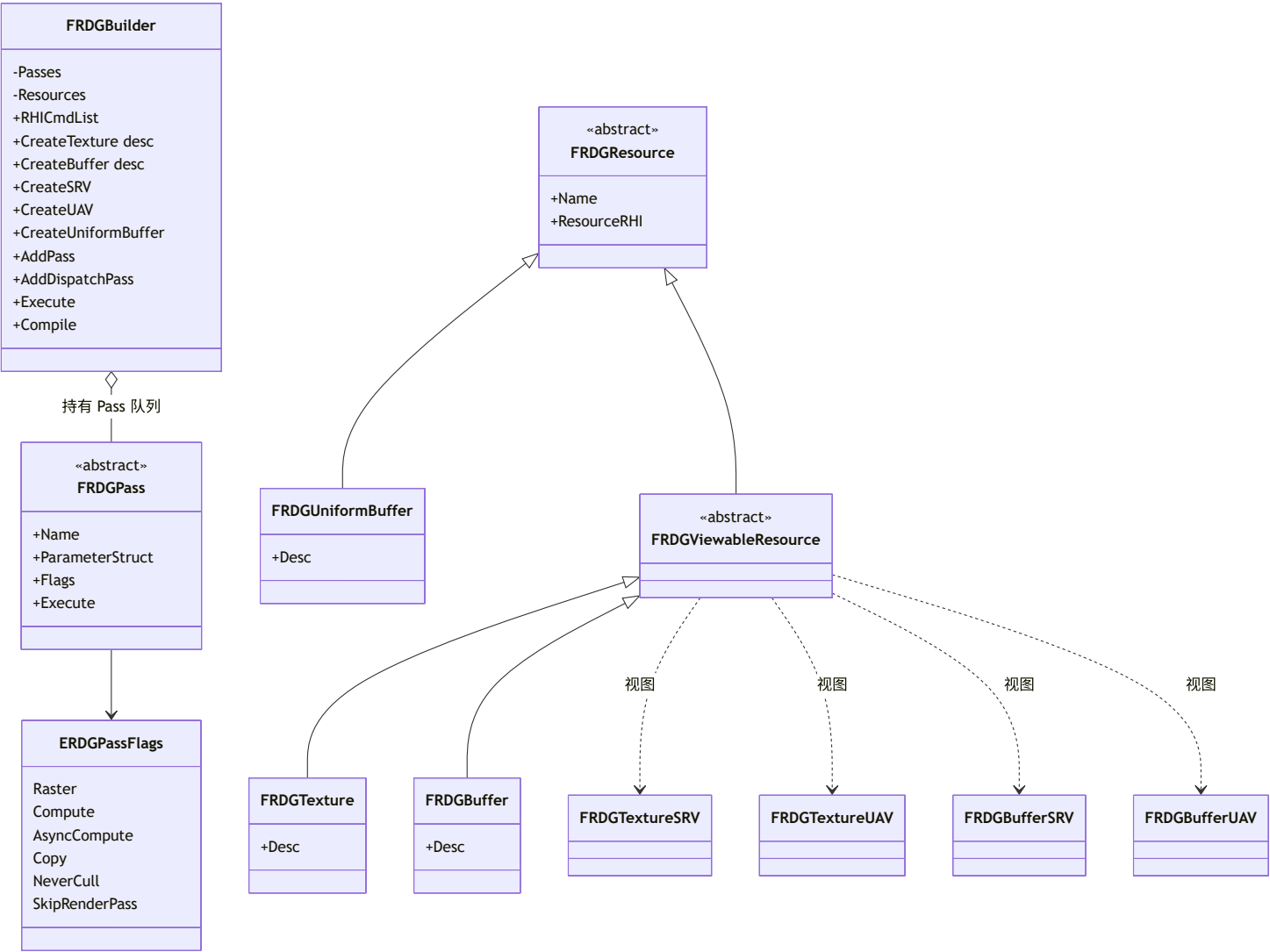
4.3 FDeferredShadingSceneRenderer::Render 全阶段



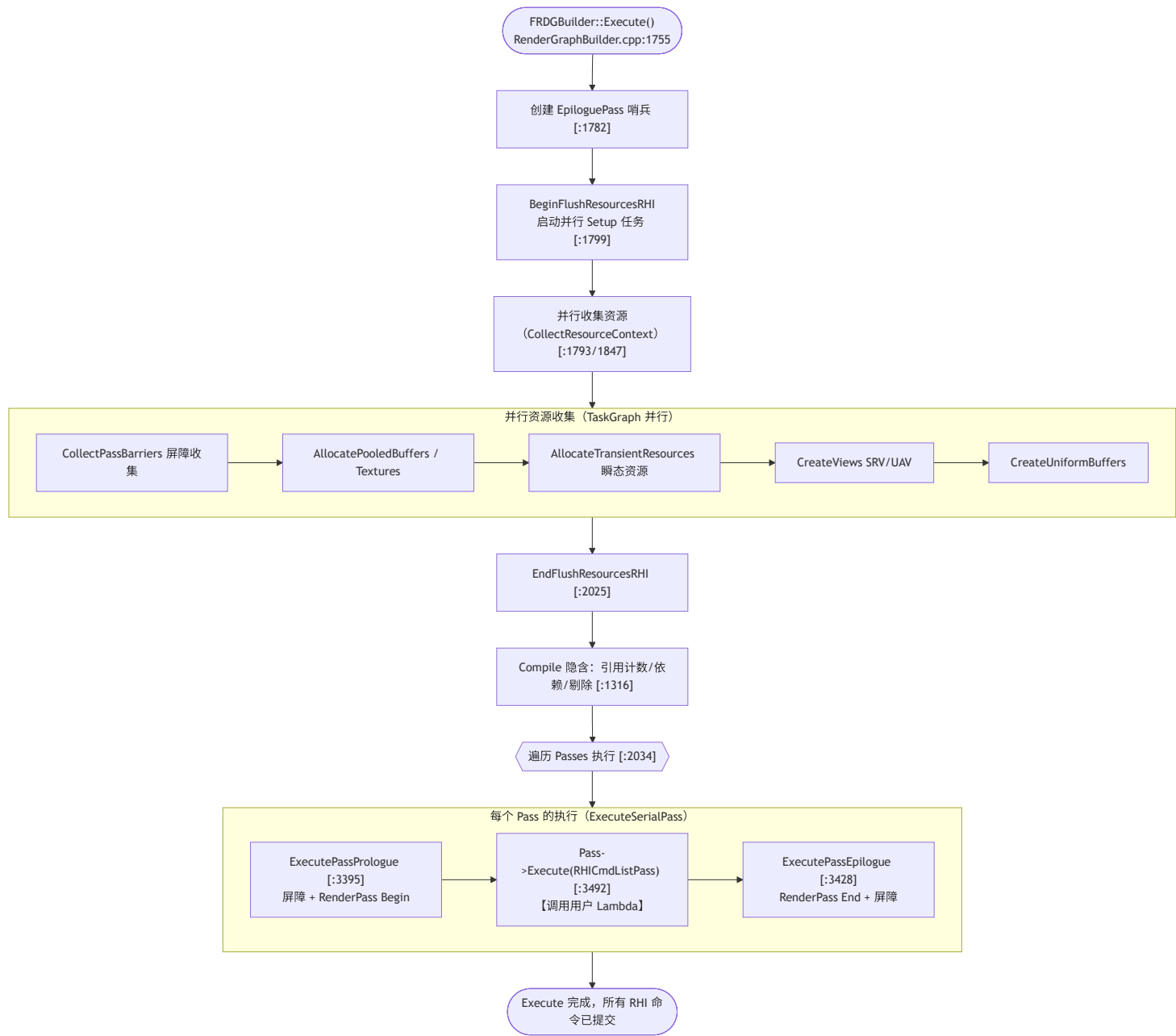




4.4 RDG 系统类图



4.5 RDG Execute 内部流程



4.6 渲染入口与 Renderer 选择

函数/类	文件
FRendererModule::BeginRenderingViewFamilies	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L5
RenderViewFamily_RenderThread	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L4
FSceneRendererBase	[SceneRendering.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.h#L202
FSceneRenderer	[SceneRendering.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.h#L207
FSceneRenderer::Render (纯虚)	[SceneRendering.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.h#L220

函数/类	文件
FDeferredShadingSceneRenderer	[DeferredShadingRenderer.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/DeferredShadingRender
FMobileSceneRenderers	[SceneRendering.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.h#L28C
FSceneRenderBuilder::CreateSceneRenderers	[SceneRenderBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRenderBuilder.cpp
FSceneRenderBuilder::Execute 内创建 GraphBuilder	[SceneRenderBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRenderBuilder.cpp
GraphBuilder.Execute() 提交点	[SceneRenderBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRenderBuilder.cpp

4.7 FDeferredShadingSceneRenderers::Render 关键步骤

阶段	文件
Render	[DeferredShadingRenderer.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/DeferredShadingRenderer.cpp#L1736)
OnRenderBegin	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L3913)
FMobileSceneRenderers::Render	[MobileShadingRenderer.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/MobileShadingRenderer.cpp#L1027)

4.8 InitViews 阶段

函数	文件
FDeferredShadingSceneRenderers::BeginInitViews	[SceneVisibility.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneVisibility.cpp#L5
FDeferredShadingSceneRenderers::EndInitViews	[SceneVisibility.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneVisibility.cpp#L6
FMobileSceneRenderers::InitViews	[MobileShadingRenderer.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/MobileShadingRender
LaunchVisibilityTasks	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#

4.9 RDG 核心（FRDGBuilder）

重要：RDG 实现位于 Runtime\RenderCore （不是 RHI）。

成员	文件	行号
class FRDGBuilder	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L47)	47
构造函数（持 RHICmdList 引用）	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L575)	575-580
CreateTexture	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L101)	101
CreateBuffer	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L107)	107

成员	文件	行号
CreateUniformBuffer	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L147)	147
AddPass （3 重载）	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L218)	218/222
AddDispatchPass	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L237)	237
AddPassDependency	[RenderGraphBuilder.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphBuilder.h#L246)	246
Execute	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L1755)	1755
Compile	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L1316)	1316
ExecutePass	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L3482)	3482-34
ExecuteSerialPass	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L3497)	3497-35
ExecutePassPrologue / Epilogue	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L3395)	3395 / 3
SetupPassDependencies	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L2319)	2319
CompilePassOps	[RenderGraphBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L2658)	2658

4.10 RDG 资源与 Pass 类

类	文件
FRDGResource	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L130)
FRDGTexture	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L569)
FRDGBuffer	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L1319)
FRDGUniformBuffer	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L191)
FRDGTextureSRV / UAV	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L837)
FRDGBufferSRV / UAV	[RenderGraphResources.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphResources.h#L1424)
class FRDGPass	[RenderGraphPass.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphPass.h#L216)
ERDGPassFlags	[RenderGraphPass.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphPass.h)
FRDGPass::Execute （模板子类）	[RenderGraphPass.h] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Public/RenderGraphPass.h#L694)

4.11 关键 RenderPass 实现

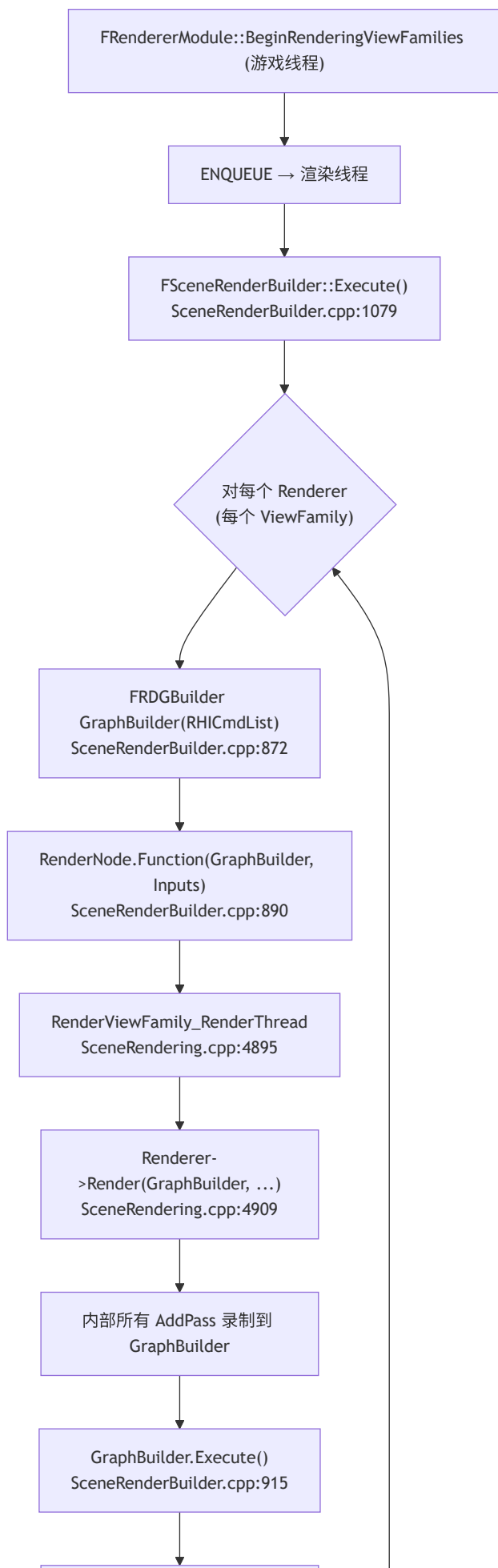
Pass	函数	文件
GBuffer / BasePass	FDeferredShadingSceneRenderer::RenderBasePass	[BasePassRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
BasePass 内部	RenderBasePassInternal	[BasePassRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
Shadow	FSceneRenderer::RenderShadowDepthMaps	[ShadowDepthRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
Lumen GI	RenderLumenSceneLighting	[LumenSceneLighting.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
Velocity	FSceneRenderer::RenderVelocities	[VelocityRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
Translucency	FDeferredShadingSceneRenderer::RenderTranslucency	[TranslucentRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
Translucency 内部	RenderTranslucencyInner	[TranslucentRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
半透明光照体素	RenderTranslucencyLightingVolume	[TranslucentLighting.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)
PostProcessing	AddPostProcessingPasses	[PostProcessing.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/Rendering/DeferredShadingSceneRenderer.cpp)

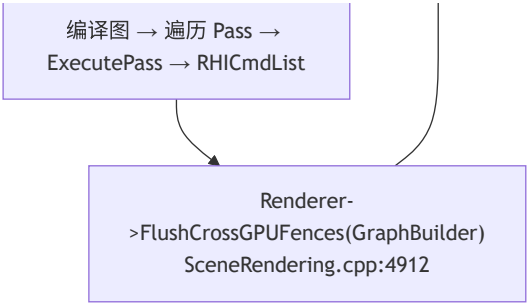
4.12 Pass 在 Render 函数中的精确调用顺序

顺序	行号	调用
1	2316	EndInitViews(GraphBuilder, LumenFrameTemporaries, InstanceCullingManager, ...)
2	2396	RenderVelocities(...) (EarlyZPass 时 Opaque 速度)
3	2823 / 2849	RenderShadowDepthMaps(...) (Forward 或可配置提前)
4	2859	CompositionLighting.ProcessBeforeBasePass (DBuffer / Decal)
5	2899	RenderLumenSceneLighting(...)
6	2905	RenderBasePass(...) (GBuffer)
7	3130	RenderShadowDepthMaps(...) (延迟阴影, 未提前时)
8	3213	RenderVelocities(...) (Opaque 速度, BasePass 未输出)
9	3265	RenderDiffuseIndirectAndAmbientOcclusion(...)
10	3314	RenderLights(...) (延迟光照)
11	3326	RenderTranslucencyLightingVolume(...)
12	3448	RenderLightShaftOcclusion(...)
13	3466	RenderFog(...)
14	3520 / 3654	RenderTranslucency(...) (UnderWater / AboveWater)
15	3638	RenderLumenFrontLayerTranslucencyReflections(...)
16	3676	RenderVelocities(...) (Translucent 速度)
17	3860-3943	AddPostProcessingPasses(...) (后处理)

4.13 RHI 提交链路

整个帧的 RHI 命令流：





项	文件
每个 Renderer 的 GraphBuilder.Execute	[SceneRenderBuilder.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRenderBuilder.cpp#L915)
FSceneRenderer::FlushCrossGPUFences	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L3822)
CleanupViewFamilies_RenderThread	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L4915)
PerFrameCleanupIfSkipRenderer	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L5251)
渲染线程唤醒	[SceneRendering.cpp] (file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRendering.cpp#L5221)

5. 完整调用链总览

5.1 启动阶段（一次性）

```
WinMain (LaunchWindows.cpp:315)
└─ LaunchWindowsStartup (185)
  └─ GuardedMainWrapper (113)
    └─ GuardedMain (Launch.cpp:87)
      └─ EnginePreInit → GEngineLoop.PreInit (LaunchEngineLoop.cpp:4332)
        └─ PreInitPreStartupScreen (1699)
          └─ LoadCoreModules (4361) ← 加载 CoreUObject
          └─ IProjectManager::LoadProjectFile (2478) ← 解析 .uproject
          └─ FTaskGraphInterface::Startup (2552)
          └─ LoadPreInitModules (4379) ← Engine/Renderer/RenderCore
          └─ LoadStartupCoreModules (4438)
          └─ PreInitPostStartupScreen (3387)
            └─ RHIIInit (3111) ← 创建 GDynamicRHI
            └─ GShaderCompilingManager = new (3191)
            └─ InitializeShaderTypes (3235)
            └─ CompileGlobalShaderMap (3247)
            └─ LoadStartupModules (4604)
            └─ InitRenderingThread ← 启动渲染线程
          └─ EngineInit → GEngineLoop.Init (4682)
            └─ GEngine = NewObject<UEngine> (4714)
            └─ GEngine->Init(this) (4763)
              └─ UGameEngine::Init (GameEngine.cpp:1205)
                └─ UEngine::Init (UnrealEngine.cpp:2052)
                  └─ GameInstance = NewObject<UGameInstance> (1249)
                    └─ InitializeStandalone → CreateNewWorldContext
                    └─ 创建 GameViewportClient (1274)
              └─ GEngine->Start() (4811)
                └─ GameInstance->StartGameInstance → 加载默认地图
            └─ while(!Exit) EngineTick → GEngineLoop.Tick (5536)
```


5.2 每帧阶段（循环）

```
FEngineLoop::Tick [LaunchEngineLoop.cpp:5536]
├─ UpdateTimeAndHandleMaxTickRate [5682]
├─ RHI BeginFrame (ENQUEUE_RENDER_COMMAND) [5689]
├─ FScene::StartFrame [5694]
├─ ENQUEUE ResetDeferredUpdates [5740]
├─ PumpMessages / 输入轮询 [5753, 5790]
├─ GEngine->Tick(...) [5828]
├─ UGameEngine::Tick [GameEngine.cpp:1740]
├─   for each FWorldContext [1860]
├─     TickWorldTravel [1872]
├─     Context.World()->Tick(...) [1880]
├─       UWorld::Tick [LevelTick.cpp:1477]
├─         SetupPhysicsTickFunctions [1711]
├─         TickTaskManager.StartFrame [1713]
├─         RunTickGroup(TG_PrePhysics) [1721]
├─         RunTickGroup(TG_StartPhysics) [1730]
├─         RunTickGroup(TG_DuringPhysics, false) [1736] ← 不阻塞
├─         RunTickGroup(TG_EndPhysics) [1743]
├─         RunTickGroup(TG_PostPhysics) [1749]
├─         TimerManager / FTickableGameObject [1787/1792]
├─         StartAsyncSendAllEndOfFrameUpdates [1797]
├─           SendAllEndOfFrameUpdates → DoDeferredRenderUpdates
├─             ENQUEUE_RENDER_COMMAND(UpdateTransform/Add/Remove)
├─         RunTickGroup(TG_PostUpdateWork) [1848]
├─         RunTickGroup(TG_LastDemotable) [1854]
├─         TickTaskManager.EndFrame [1857]
├─ FTickableGameObject::TickObjects [1947]
├─ GameViewport->Tick [1977]
├─ RedrawViewports → Viewport->Draw [2005 → 775 → 789]
├─   【GT→RT】触发一帧渲染：
├─     BeginRenderingViewFamilies (SceneRendering.cpp:5039)
├─     SceneRenderBuilder 创建 + 选择 Renderer
├─     ENQUEUE → RenderThread:
├─       FSceneRenderBuilder::Execute
├─       new FRDGBuilder(RHICmdList)
├─       Renderer->Render(GraphBuilder, ...) [DeferredShadingRenderer.cpp:1736]
├─         OnRenderBegin [1790]
├─         BeginInitViews / EndInitViews [2052/2316]
├─         RenderBasePass (GBuffer) [2905]
├─         RenderLights [3314]
├─         RenderTranslucency [3520]
├─         AddPostProcessingPasses [3860]
├─       GraphBuilder.Execute() [915]
├─       编译 → 资源分配 → 逐 Pass ExecutePass → RHICmdList
├─ Streaming / Audio 更新 [2062/2070]
├─ FSlateApplication::Tick [5890/5960]
├─ FTSTicker / FThreadManager [6071/6072]
├─ GEngine->TickDeferredCommands [6077]
├─ FCoreDelegates::OnEndFrame [6095]
├─ RHI EndFrame (ENQUEUE_RENDER_COMMAND) [6131]
```

6. 总结要点

1. **初始化顺序**：RHI 初始化（RHIInit）位于 **PreInit 阶段**（不是 Init 阶段），在 Shader 系统初始化之前完成。GEngine 在 Init 阶段创建。
2. **TickGroup 因果性**：TG_DuringPhysics 是唯一用 bBlockTillComplete=false 调用的组，允许 GameThread 与物理并行跑；后续 TG_EndPhysics 会阻塞等待。
3. **SceneProxy 是镜像**：FPrimitiveSceneProxy 是 UPrimitiveComponent 在渲染线程的**只读镜像**。GT 只能通过 ENQUEUE_RENDER_COMMAND 单向通信修改 RT 状态。

4. **整场景刷新**：所有增删改命令进入 `FScenePrimitiveUpdates` 队列，由 `FScene::Update` ([`RendererScene.cpp:5244`] (`file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp#L5244`)) 在下一次渲染前一次性消费。
5. **RDG 在 RenderCore**：RDG 核心位于 `Runtime\RenderCore` (不是 RHI)。 `FRDGBuilder::Execute` ([`RenderGraphBuilder.cpp:1755`] (`file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderGraphBuilder.cpp#L1755`)) 完成「编译 → 资源分配 → 逐 Pass 执行」。
6. **每个 ViewFamily 一个 RDG**： `SceneRenderBuilder.cpp:872` 创建独立 `FRDGBuilder`，行 915 执行——这意味着每个 `ViewFamily` 有独立的依赖图，而非整帧一个图。
7. **渲染器选择**： [`SceneRenderBuilder.cpp:496-519`] (`file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/SceneRenderBuilder.cpp#L496`) 通过 `GetFeatureLevelShadingPath(FeatureLevel)` 决定 `Deferred / Mobile`。

7. 术语对照表（用户记忆 vs 实际源码）

用户提到的名称	实际源码名称	位置
<code>FTickFunctionManager</code>	<code>FTickTaskManager</code> (+ <code>FTickTaskSequencer</code>)	[<code>TickTaskManager.cpp:1965</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L1965</code>)
<code>QueueTickFunction</code>	<code>FTickFunction::QueueTickFunction</code>	[<code>TickTaskManager.cpp:2617</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Private/TickTaskManager.cpp#L2617</code>)
<code>FDeferredUpdateInterface</code>	<code>FDeferredUpdateResource</code>	[<code>TextureResource.h:285</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Engine/Public/TextureResource.h#L285</code>)
<code>StartRenderCommandFence</code>	<code>FRenderCommandFence::BeginFence</code>	[<code>RenderingThread.cpp:978</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L978</code>)
<code>FlushRendering</code>	<code>FlushRenderingCommands</code> (无后缀版本不存在)	[<code>RenderingThread.cpp:1272</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RenderCore/Private/RenderingThread.cpp#L1272</code>)
<code>ULevel::Tick</code>	不存在 (由 <code>FTickTaskLevel</code> 驱动)	<code>TickTaskManager.cpp</code>
<code>GRHI</code>	<code>GDynamicRHI</code> + <code>GRHICommandList</code> (非单一指针)	[<code>DynamicRHI.cpp:35</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/RHI/Private/DynamicRHI.cpp#L35</code>)
<code>FScene</code> 实现位置	已从 <code>SceneRendering.cpp</code> 迁移到 <code>RendererScene.cpp</code>	[<code>RendererScene.cpp</code>] (<code>file:///j:/UnrealSource/UnrealEngine/Engine/Source/Runtime/Renderer/Private/RendererScene.cpp</code>)